Exercise 1.4

Q1 Let $P(x)$ denote the statement "$x \leq 4$". What are these truth values?

(a) $P(0)$ — True
(b) $P(4)$ — True
(c) $P(6)$ — False

Q2 Let $P(x)$ be the statement "The word x contains the letter a". What are these truth values?

(a) P(orange) — True
(b) P(lemon) — False
(c) P(true) — False
(d) P(false) — True

Q3 Let $Q(x, y)$ denote the statement "x is the capital of y". What are these truth values?

(a) Q(Denver, Colorado) — True
(b) Q(Detroit, Michigan) — False
(c) Q(Masachusetts, Boston) — False
(d) Q(New York, New York) — False

Q4 State the value of x after the statement if $P(x)$ then $x := 1$ is executed, where $P(x)$ is the statement "$x > 1$", if the value of x when this statement is reached

(a) $x = 0$ — False
(b) $x = 1$ — False
(c) $x = 2$ — True

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Q5 let $P(x)$ be the statement "x spends more than five hours every weekday in class", where the domain for x consists of all students. Express each of these quantifiers in English.

(a) $\exists x P(x)$

There are some students spends more than five hours every weekday in class.

(b) $\forall x (P(x))$

All students spend more than five hours every weekday in class.

(c) $\exists x \neg (P(x))$

Some students do not spend more than 5 hours every weekday in class.

(d) $\forall x \neg P(x)$

No student spend more than 5 hours every weekday in class.

Q6 let $N(x)$ be the statement "x has visited North Dakota", where the domain of the students in your school. Express each of these quantifications in English.

(a) $\exists x N(x)$

There is one student in your school who has visited North Dakota.

(b) $\forall x (N x)$

All students in your school who have visited North Dakota.

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(c) $\neg\exists x N(x)$

There is no student who has visited North Dakota.

(d) $\exists x \neg N(x)$

There is a student who has not visited North Dakota.

(f) $\neg \forall x N(x)$

No student in your school who has visited North Dakota.

(e) $\forall x \neg N(x)$

Every student in your school who have not visited North Dakota.

Q7 Translate these statements into English where $C(x)$ is "x is a comedian" and $F(x)$ is "x is funny" and domain consists of all people.

(a) $\forall x (C(x) \rightarrow F(x))$

All people are comedian and funny.

(b) $\forall x (C(x) \wedge F(x))$

All people are comedian and funny.

(c) $\exists x (C(x) \rightarrow F(x))$

There is a comedian who is funny.

(d) $\exists x (C(x) \wedge F(x))$

Some comedian are funny.

Q8 Translate these statements into English, where $R(x)$ is "x is a rabbit" and $H(x)$ is "x

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hops", and domain consists of all animals.

(a) $\forall x(R(x) \rightarrow H(x))$
All animals are rabbits and hops.

(b) $\forall x(R(x) \wedge H(x))$
All animals are rabbits and hops.

(c) $\exists x(R(x) \rightarrow H(x))$
There is a rabbit and ~~who~~ is hop.

(d) $\exists x(R(x) \wedge H(x))$
There are some rabbits and hops.

Q9. Let $P(x)$ be the statement "x can speak Russian" and $Q(x)$ "x knows the computer language C++".

(a) There is a student at your school who can speak Russian and who knows C++.
$\forall \exists x(P(x) \wedge Q(x))$

(b) There is a student at your school who can speak Russian but does not know C++.
$\exists x(P(x) \wedge \neg Q(x))$

(c) Every student either can speak Russian or know C++.
$\forall x(P(x) \vee Q(x))$

(d) No student at your school can speak Russian or knows C++.
$\neg \forall x(P(x) \vee Q(x))$

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Q10 Let C(x) be statement "x has a cat", D(x) "x has a dong" F(x) "x has a ferret", domain consist of all students.

(a) A student has a cat, a dong & ferrit.
$\exists x (C(x) \wedge D(x) \wedge F(x))$

(b) All students have a cat, dog or ferrit.
$\forall x (C(x) \vee D(x) \vee F(x))$

(c) Some students has a cat and ferrit but not a dog.
$\exists x (C(x) \wedge F(x) \wedge \neg D(x))$

(d) For each animal cat, dog and ferrit there is a student who has this animal as pet.
$\exists x (C(x) \wedge D(x) \wedge F(x))$

(e) No student in your school has a cat, dogs and ferrit.
$\neg \forall x (C(x) \wedge D(x) \wedge F(x))$

Q11 Let P(x) be the statement "$x = x^2$". If domain consists of Integers, what are these truth values?

(a) P(0) — True
(b) P(1) — True
(c) P(2) — False
(d) P(-1) — False
(e) $\exists x P(x)$ — True
(f) $\forall x P(x)$ — False.

Q12 Let Q(x) be the statement "$x+1 > 2x$" if domain consists of all integers, what

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are these truth values?

(a) $Q(0)$ since $0+1>2(0) \Rightarrow 1>0$, so it is true
(b) $Q(-1)$ $-1+1>2(-1) \Rightarrow 0>-2$, it is true
(c) $Q(1)$ $1+1>2(1) \Rightarrow 2>2$, it is false
(d) $\exists x Q(x)$ from (a) we know that there is at least one x
(e) $\forall x Q(x)$ from (c) we known that there is at least one x that makes $Q(x)$ false, it is false
(f) $\exists x \neg Q(x)$ from (c) we know that there is at least one x that makes Qx false, so it is true.
(g) $\forall x \neg Q(x)$ from (b) we know that there is at least one x that makes $Q(x)$ true, so it is false.

Q13 Determine the truth value of each of these statements if domain consists of all integers.

(a) $\forall n(n+1>n)$ True
(b) $\exists n(2n=3n)$ True
(c) $\exists x(n = -n)$ True
(d) $\forall x(3n \leq 4n)$ True

Q14 Determin the truth value of each of these statements if domain consists of all real numbers.

(a) $\exists x(x^3=-1)$ $(-1)^3=-1$, this is true
(b) $\exists x(x^4<x^2)$ $(\frac{1}{2})^4<(\frac{1}{2})^2$, this is true
(c) $\forall x((-x^2)^2=x^2)$ $(-x)^2=((-1)x)^2=(-1)^2x^2=x^2$, it is true
(d) $\forall x(2x>x)$ Twice a positive number is larger than number, but the inequality is not true for negative or 0, it is false.

Q15 Determine the truth value of these statements if domain for all variables consists of all integers.

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(a) $\forall n(n^2 \geq 0)$ True

(b) $\exists n(n^2 = 2)$ False

(c) $\forall n(n^2 \geq n)$ True

(d) $\exists n(n^2 < 0)$ False

Determine truth value of each of these statements if domain of each variable consists of all real numbers.

(a) $\exists x(x^2 = 2)$ true $(x = \sqrt{2})$

(b) $\exists(x)(x^2 = -1)$ false ($\sqrt{-1}$ is no real number)

(c) $\forall x(x^2 + 2 \geq 1)$ true (as left side is always at least 2)

(d) $\forall x(x^2 \neq x)$ false (not true for $x=1$ or $x=0$)

Suppose that domain of propositional function $P(x)$ consists of integers 0, 1, 2, 3, 4. Write out each of these propositions using disjunctions, conjunctions, negation.

(a) $\exists x\, P(x)$

$P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$

(b) $\forall x\, P(x)$

$P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$

(c) $\exists x \neg P(x)$

$\neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$

(d) $\forall x \neg P(x)$

$\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$

(e) $\neg \exists x\, P(x)$

$\neg(P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4))$

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(f) $\neg \forall x P(x)$

$\neg(P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4))$

Q18 Suppose that the domain of propositional function $P(x)$ consists of integers $-2, -1, 0, 1$ and 2.

(a) $\exists x P(x)$

$P(-2) \vee P(-1) \vee P(0) \vee P(1) \vee P(2)$

(b) $\forall x P(x)$

$P(-2) \wedge P(-1) \wedge P(0) \wedge P(1) \wedge P(2)$

(c) $\exists x \neg P(x)$

$\neg P(-2) \vee \neg P(-1) \vee \neg P(0) \vee \neg P(1) \vee \neg P(2)$

(d) $\forall x \neg P(x)$

$\neg P(-2) \wedge \neg P(-1) \wedge \neg P(0) \wedge \neg P(1) \wedge \neg P(2)$

(e) $\neg \exists x P(x)$

$\neg(P(-2) \vee P(-1) \vee P(0) \vee P(1) \vee P(2))$

(f) $\neg \forall x P(x)$

$\neg(P(-2) \wedge P(-1) \wedge P(0) \wedge P(1) \wedge P(2))$

Q19 Suppose that domain of propositional function $P(x)$ consists of integer $1, 2, 3, 4, 5$. Express these statements without using quantifiers, instead using only negation, disjunctions, conjunctions.

(a) $\exists x P(x)$

$(P(1) \vee P(2) \vee P(3) \vee P(4) \vee P(5)$

(b) $\forall x P(x)$

$P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5)$

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(c) $\neg \exists x P(x)$

$\neg (P(1) \vee P(2) \vee P(3) \vee P(4) \vee P(5))$

(d) $\neg \forall x P(x)$

$\neg (P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5))$

(e) $\forall x((x \neq 3) \rightarrow P(x)) \vee \exists x \neg P(x)$

$(P(1) \wedge P(2) \wedge P(4) \wedge P(5)) \vee (\neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4) \vee \neg P(5))$

Suppose that the domain of propositional function $P(x)$ consists of $-5, -3, -1, 1, 3, 5$. Express these statements without using quantifiers, instead using only negations, disjunctions, conjunctions.

(a) $\exists x P(x)$

$P(-5) \vee P(-3) \vee P(-1) \vee P(1) \vee P(3) \vee P(5)$

(b) $\forall x P(x)$

$P(-5) \wedge P(-3) \wedge P(-1) \wedge P(1) \wedge P(3) \wedge P(5)$

(c) $\forall x((x \neq 1 \rightarrow P(x))$

$P(-5) \wedge P(-3) \wedge P(-1) \wedge P(3) \wedge P(5)$

(d) $\exists x((x \geq 0) \wedge P(x))$

$(P(1) \vee P(3) \vee P(3)) \wedge (P(-5) \wedge P$

(e) $\exists x(\neg P(x)) \wedge \forall x((x < 0) \rightarrow P(x))$

For each of these statements find a domain for which the statement is true and a domain for which the statement is false.

(a) Everyone is studying discrete math.

All student in your discrete class studing discrete, true all students in the world; False.

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(b) Everyone is older than 21 years.
MSC girl are older than 21; BS student are not older than 21.

(c) Every two people have the same mother.
Sahar and Fatir have same mother; Everyone has same mother;

(d) No two different people have the same grandmother.
Sahar and yours ; All politicians;

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23: Translate in two ways each of these statements into logical expressions using predicates, quantifiers, logical connectives. First let domain consist of students in your class and second, let it consists of all people.

In order to do translation, the second way, we let C(x) be the propositional function "x is in your class". Note that for second ways, we always want to use conditional statements with universal quantifiers and conjunctions with existantial quantifiers.

a) Someone is your class can speak hindi.

Let H(x) be "x can speak hindi", Then we have $\exists x H(x)$ the first way or $\exists x (C(x) \wedge H(x)$ the second way.

b) Everyone in your class is friendly.

i) $\exists x H(x)$

ii) $\exists x (C(x) \wedge H(x))$

c) There is a person in your class who was not born in California.

i) $\exists x \neg B(x)$

ii) $\exists x (C(x) \wedge \neg B(x))$

d) A student in your class has been in movie?

i) $\exists x M(x)$

ii) $\exists x (C(x) \wedge M(x))$

e) No student in your class has taken a course in logic programming.

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i) $\forall x \neg L(x)$

ii) $\forall x (C(x) \wedge \neg L(x))$

24. Translate: First, let the domain consists of students in your class, second let it consists of all people.

a) Everyone in your class has a cellular phone

i) $\forall x P(x)$

ii) $\forall x (C(x) \rightarrow P(x))$

b) Somebody in your class has seen a foreign movie.

i) $\exists x F(x)$

ii) $\exists x (C(x) \wedge F(x))$

c) There is a person in your class who can't swim.

i) $\exists x \neg S(x)$

ii) $\exists x (C(x) \wedge \neg S(x))$

d) All students in your class can solve quadratic equations.

i) $\forall x Q(x)$

ii) $\forall x (C(x) \wedge Q(x))$

e) Some students in your class ~~dose solve~~ not want to be rich.

i) $\exists x \neg R(x)$

ii) $\exists x (C(x) \wedge \neg R(x))$

25. Translate:

a) No one is perfect.

$\forall x \neg (P(x)$

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b) Not everyone is perfect.
$\neg \forall x P(x)$

c) All your friends are perfect.
$\forall x (F(x) \rightarrow P(x))$

d) At least one of your friends is perfect.
$\exists x (F(x) \wedge P(x))$

e) Everyone is your friend and is perfect.
$\forall x (F(x) \wedge P(x))$

f) Not everybody is your friend or someone is not perfect.
$(\neg \forall x F(x)) \vee (\exists x \neg P(x))$

26: Translate.

a) Someone in your school has visited Uzbekistan.
$\exists x (S(x) \wedge U(x))$

b) Everyone in your class has studied calculus and C++.
$\forall x (C(x) \wedge P(x))$

c) No one in your class school owns both a bicycle and motorcycle.
$\forall x (S(x) \rightarrow \neg B(x) \wedge M(x))$

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d) There is a person in your school who is not happy.

$\exists x (S(x) \wedge \neg H(x))$

e) Everyone in your school was born in 12th century.

$\forall x (S(x) \rightarrow T(x))$

30) Suppose the domain of propositional function $P(x, y)$, x is 1, 2 or 3, y is 1, 2 or 3. Write out these propositions using conjunction and disjunction.

a) $\exists x P(x, 3)$

$P(1,3) \vee P(2,3) \vee P(3,3)$

b) $\forall y P(1, y)$

$P(1,1) \wedge P(1,2) \wedge P(1,3)$

c) $\exists x \neg P(2, y)$

$\neg P(2,1) \vee \neg P(2,2) \vee \neg P(2,3)$

d) $\forall x \neg P(x, 2)$

$\neg P(1,2) \wedge \neg P(2,2) \wedge \neg P(3,2)$

31) Suppose that domain $Q(x, y, z)$ $x = 0, 1$ or 2, $y = 0$ or 1, $z = 0$ or 1

a) $\forall y Q(0, y, 0)$

$Q(0,0,0) \wedge Q(0,1,0)$

b) $\exists x Q(x, 1, 1)$

$Q(0,1,1) \vee Q(1,1,1) \vee Q(2,1,1)$

c) $\exists z \neg Q(0,0,z)$

$\neg Q(0,0,0) \vee \neg Q(0,0,1)$

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d) $\exists x \neg Q(x, 0, 1)$

$\neg Q(0,0,1) \vee \neg Q(1,0,1) \vee \neg Q(2,0,1)$

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